CRITICAL CARE MEDICINE · MIDTERM · LECTURE 1 · QUESTION BANK

Acute Coronary Syndrome (ACS)

متلازمة الشريان التاجي الحادة — بنك الأسئلة

39	12	15
Past exam questions	NCLEX-style	Flashcards

المصادر اللي اتجمعت منها أسئلة المحاضرة دي:

كويزات سابقة (طب حالات حرجة) · أسئلة سابقة

جنب كل سؤال مصدره. لو السؤال اتكرر في كذا امتحان مكتوب كل مرة بمصدرها، وعليه علامة "تكرر". الإجابات كلها في آخر الملف.

اللَّهُمَّ لا سَهْلَ إِلَّا ما جَعَلْتَهُ سَهْلًا، وأنتَ تَجْعَلُ الحَزْنَ إذا شِئْتَ سَهْلًا — صلِّ على النبي ﷺ

Part A · Past Exam Questions

أسئلة الدفعات السابقة

1 Acute Coronary Syndrome term includes STEMI and Non STEMI only. — كويزات سابقة (طب حالات حرجة)

a. True **b.** False

2 The definitive treatment for patient with ST-elevation Myocardial Infarction (STEMI) is ..... — كويزات سابقة (طب حالات حرجة)

a. Oxygen
b. Nitrates
c. Coronary angiography and coronary revascularization
d. Good hydration

3 Treatment of Acute Coronary Syndrome doesn't include: — كويزات سابقة (طب حالات حرجة)

a. Antiplatelet drugs **b.** Nitrates
c. Beta blocker drugs **d.** Aminophylline

4 Differentiation between Myocardial Infarction and Angina could be done through: — كويزات سابقة (طب حالات حرجة)

a. Chest pain characteristics **b.** Checking vital signs
c. Checking cardiac enzymes **d.** Identification of risk factors

5 The definitive treatment for patient with ST elevation Myocardial Infarction (STEMI) is: — كويزات سابقة (طب حالات حرجة)

a. Coronary angiography & revascularization
b. Good hydration
c. Oxygen
d. Nitrates

6 **Management of acute ST elevation MI includes morphine, oxygen, nitrates only.** كويزات سابقة (طب حالات حرجة)

a. True **b.** False

7 **GRACE ACS risk includes all of the following except:** كويزات سابقة (طب حالات حرجة)

a. Age **b.** Heart rate

c. Sodium concentration **d.** Systolic blood pressure

8 **Differentiation between Myocardial Infarction and Angina could be done through checking:** كويزات سابقة (طب حالات حرجة)

a. Vital signs **b.** Pain characteristics

c. Cardiac enzymes **d.** Risk factors

9 **What does Acute Coronary Syndrome (ACS) encompass?** أسئلة سابقة

a. Only unstable angina

b. A spectrum of conditions including unstable angina, NSTEMI, and STEMI

c. Only ST-Elevation Myocardial Infarction (STEMI)

d. A condition only seen in older adults

10 **Which of the following is the hallmark symptom of Acute Coronary Syndrome?** أسئلة سابقة

a. Abdominal pain

b. Chest pain, often described as pressure, tightness, or heaviness

c. Dizziness

d. Hemorrhage

11 **What laboratory test is most useful in diagnosing ACS?** أسئلة سابقة

a. Complete Blood Count (CBC) **b.** Cardiac Enzymes (e.g., Troponin)

c. Liver Function Tests **d.** Arterial Blood Gas (ABG)

12 **What ECG change is commonly seen in Unstable Angina (UA) and NSTEMI?** أسئلة سابقة

a. ST segment elevation

b. T-wave inversion or ST segment depression

c. Q waves

d. No changes

13 **What is the main difference between NSTEMI and STEMI?** أسئلة سابقة

a. NSTEMI involves complete occlusion of a coronary artery, whereas STEMI does not

b. NSTEMI involves partial occlusion of a coronary artery, whereas STEMI involves complete occlusion

c. STEMI is less severe than NSTEMI

d. STEMI does not require any treatment

14 Which of the following is a typical trigger for chest pain in ACS? أسئلة سابقة

a. Physical activity, emotional stress, or cold weather
b. Sleeping
c. Eating meals
d. High temperature

15 In ACS, which of the following is used to relieve chest pain? أسئلة سابقة

a. Exercise
b. Nitroglycerin
c. Corticosteroids
d. Antihistamines

16 What does the TIMI (Thrombolysis in Myocardial Infarction) Risk Score help determine? أسئلة سابقة

a. The presence of ST-segment elevation
b. The prognosis of death and ischemic events in ACS patients
c. The exact location of plaque rupture
d. The number of coronary arteries involved

17 What does the GRACE Risk Model in ACS assess? أسئلة سابقة

a. The likelihood of developing hypertension
b. The risk of in-hospital mortality
c. The severity of atherosclerotic plaques
d. The occurrence of arrhythmias

18 Which of the following is a common symptom of STEMI? أسئلة سابقة

a. Mild chest discomfort
b. Severe chest pain with radiation, nausea, and sweating
c. Shortness of breath only
d. Pain that is relieved by rest

19 Which of the following ECG changes is commonly seen in STEMI? أسئلة سابقة

a. Significant ST-segment elevation in ≥2 contiguous leads
b. Q waves without ST-segment changes
c. T-wave flattening only
d. ST-segment depression only

20 What does elevated troponin levels indicate in the context of ACS? أسئلة سابقة

a. Dehydration
b. Myocardial necrosis
c. Hyperglycemia
d. Kidney failure

21 What is the key feature distinguishing Unstable Angina (UA) from Stable Angina? أسئلة سابقة

a. The pain occurs at rest or with minimal exertion in UA
b. Stable angina is relieved by nitroglycerin, while UA is not
c. UA is not associated with pain
d. Stable angina does not cause chest pain

22 **Which treatment is part of the initial management of ACS?** أسئلة سابقة

a. Beta-blockers only
b. MONA (Morphine, Oxygen, Nitroglycerin, Antithrombotic therapy)
c. Corticosteroids
d. Sodium bicarbonate

23 **What is the purpose of nitroglycerin in ACS management?** أسئلة سابقة

a. To increase blood pressure
b. To reduce myocardial oxygen demand
c. To prevent arrhythmias
d. To dissolve blood clots

24 **Which of the following is an important part of interventional management for ACS?** أسئلة سابقة

a. Coronary Angiography followed by PCI (Percutaneous Coronary Intervention)
b. Long-term bed rest
c. Administration of corticosteroids
d. Blood transfusion

25 **Which surgical intervention is commonly used for severe cases of coronary artery disease?** أسئلة سابقة

a. Coronary Artery Bypass Grafting (CABG)
b. Stent placement only
c. Percutaneous balloon angioplasty
d. Valve replacement surgery

26 **Which of the following mechanical assist devices is used for hemodynamic support in severe cases of ACS?** أسئلة سابقة

a. V-A ECMO
b. Ventricular assist device (VAD)
c. Intra-aortic balloon pump (IABP)
d. Continuous positive airway pressure (CPAP)

27 **Which device is considered less invasive than ECMO and used for circulatory support during PCI?** أسئلة سابقة

a. Impella
b. IABP
c. V-A ECMO
d. Defibrillator

28 **What does the Intra-Aortic Balloon Pump (IABP) do?** أسئلة سابقة

a. Supports both heart and lung function
b. Improves blood flow to the heart and reduces workload during PCI
c. Treats respiratory failure
d. Provides long-term mechanical ventilation

29 **Which of the following is a typical ECG finding in NSTEMI?** أسئلة سابقة

a. Significant ST-segment elevation
b. No change in ST-segment
c. ST-segment depression or T-wave inversion
d. Q waves

30 What role does the Killip Class play in ACS? أسئلة سابقة

a. It is a classification of heart failure severity, which impacts mortality risk
b. It indicates the type of coronary artery blocked
c. It determines the number of PCI required
d. It is used to assess the extent of myocardial injury

31 What does the GRACE Risk Score assess in ACS patients? أسئلة سابقة

a. Risk of arrhythmias only
b. Risk of in-hospital mortality
c. Length of hospital stay
d. Response to antithrombotic therapy

32 What is the most appropriate treatment for STEMI when PCI is unavailable? أسئلة سابقة

a. Immediate coronary artery bypass grafting (CABG)
b. Thrombolytics
c. Long-term bed rest
d. Corticosteroids

33 What is the primary cause of myocardial ischemia in ACS? أسئلة سابقة

a. Hypoxia
b. Atherosclerotic plaque rupture and thrombus formation
c. Viral infection
d. High blood pressure

34 How does morphine assist in ACS management? أسئلة سابقة

a. It increases heart rate
b. It provides pain and anxiety relief
c. It prevents clot formation
d. It increases blood pressure

35 In the context of ACS, what is the primary action of aspirin? أسئلة سابقة

a. Increase platelet aggregation
b. Prevent thrombus formation by inhibiting platelet aggregation
c. Dilate blood vessels
d. Reduce blood pressure

36 What is a characteristic finding in the cardiac biomarkers of NSTEMI? أسئلة سابقة

a. Normal troponin levels
b. Elevated troponin levels
c. Normal creatine kinase levels
d. Decreased B-type natriuretic peptide levels

37 What does partial coronary artery occlusion typically result in during NSTEMI? أسئلة سابقة

a. Full-thickness myocardial infarction
b. Transient reversible ischemia
c. Total coronary artery blockage
d. No myocardial injury

38 Which of the following is a common sign of cardiogenic shock in ACS? أسئلة سابقة

a. High blood pressure
b. Low blood pressure with signs of organ dysfunction
c. Normal heart rate and rhythm
d. Lack of chest pain

39 What is the next step in management if a 12-lead ECG shows STEMI after ROSC? أسئلة سابقة

a. Administer amiodarone
b. Perform targeted temperature management
c. Proceed with coronary angiography and revascularization
d. Start advanced airway management

Part B · NCLEX-Style Practice

أسئلة على طريقة NCLEX (إضافة)

N1 A patient has chest pain at rest, ST depression on ECG, and elevated troponin. The most likely diagnosis is: NCLEX

a. Stable angina
b. Unstable angina
c. NSTEMI
d. STEMI

N2 ACS includes (select all): SELECT ALL

a. Unstable angina
b. NSTEMI
c. STEMI
d. Stable angina
e. Pericarditis

N3 The cardiac biomarker most specific for myocardial injury is: NCLEX

a. CK-total
b. Troponin I/T
c. LDH
d. Myoglobin

N4 A patient with STEMI arrives at a PCI-capable hospital. The target door-to-balloon time is: NCLEX

a. ≤ 90 minutes
b. ≤ 6 hours
c. ≤ 24 hours
d. ≤ 12 hours

N5 Before giving nitroglycerin to a patient with ACS, the nurse must check (select all): SELECT ALL

a. Systolic BP > 90 mmHg
b. No recent phosphodiesterase inhibitor (sildenafil)
c. No right ventricular infarction
d. Serum potassium
e. Blood glucose

N6 Which drug inhibits platelet aggregation and should be chewed early in suspected ACS? NCLEX

a. Aspirin 162–325 mg
b. Paracetamol
c. Morphine
d. Furosemide

N7 **Contraindications to thrombolytic therapy include (select all):** SELECT ALL

a. Previous hemorrhagic stroke
b. Active internal bleeding
c. Suspected aortic dissection
d. Age above 50 years
e. Recent major surgery or trauma

N8 **Beta-blockers in ACS work mainly by:** NCLEX

a. Increasing heart rate
b. Reducing myocardial oxygen demand
c. Dissolving the clot
d. Raising blood pressure

N9 **Killip class IV in a patient with MI means:** NCLEX

a. No heart failure
b. Basal crackles / S3
c. Frank pulmonary edema
d. Cardiogenic shock

N10 **Signs of reperfusion after thrombolysis include (select all):** SELECT ALL

a. Relief of chest pain
b. Resolution of ST elevation
c. Reperfusion arrhythmias
d. Rising ST segment
e. Increasing chest pain

N11 **Which nursing action has priority during the first hour for a patient with STEMI?** NCLEX

a. Teach about diet
b. Continuous ECG monitoring with IV access and pain relief
c. Ambulate the patient
d. Schedule an exercise test

N12 **Women, older adults and diabetics with ACS may present with:** NCLEX

a. Atypical symptoms such as dyspnea, fatigue or epigastric pain
b. Always crushing chest pain
c. Fever and rash
d. Bradycardia only

Part C · Quizlet Flashcards

كروت مراجعة سريعة

غطّي العمود الأبيض بورقة واختبر نفسك: شوف المصطلح وقول التعريف.

ACS	Spectrum due to acute myocardial ischemia: unstable angina, NSTEMI, STEMI.	**Pathophysiology**	Atherosclerotic plaque rupture → platelet aggregation → thrombus → partial or total occlusion.
Unstable angina	Chest pain at rest / new / worsening; ECG may show ST depression; troponin normal.	**NSTEMI**	Partial occlusion; ST depression or T inversion; troponin raised.
STEMI	Complete occlusion; ST elevation ≥ 2 contiguous leads (or new LBBB); troponin raised.	**Typical pain**	Retrosternal pressure > 20 min, radiating to jaw/left arm, with sweating, nausea, dyspnea; not relieved by rest.

Troponin	Rises 3–4 h, peaks ~24 h, stays raised 7–14 days — most specific marker.
Definitive STEMI treatment	Coronary angiography + PCI (door-to-balloon ≤ 90 min); thrombolysis if PCI unavailable (door-to-needle ≤ 30 min).
TIMI score	Predicts risk of death / ischemic events.
Killip class	I none · II crackles/S3 · III pulmonary edema · IV cardiogenic shock.
Not used in ACS	Aminophylline, corticosteroids.

MONA + BAH	Morphine, Oxygen (if SpO_2 < 90%), Nitrates, Aspirin + Beta-blocker, Anticoagulant (heparin), High-intensity statin.
CABG	Surgical bypass for left main or multivessel disease.
GRACE score	Predicts in-hospital mortality: age, HR, SBP, creatinine, Killip class, cardiac arrest, ST deviation, enzymes (no sodium).
Mechanical support	IABP, Impella, V-A ECMO for cardiogenic shock.

Answer Key

الإجابات

1. b — False
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

2. c — Coronary angiography and coronary revascularization
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

3. d — Aminophylline
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

4. c — Checking cardiac enzymes
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

5. a — Coronary angiography & revascularization
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

6. b — False
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

7. c — Sodium concentration
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

8. c — Cardiac enzymes
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

9. b — A spectrum of conditions including unstable angina, NSTEMI, and STEMI

10. b — Chest pain, often described as pressure, tightness, or heaviness

11. b — Cardiac Enzymes (e.g., Troponin)

12. b — T-wave inversion or ST segment depression

13. b — NSTEMI involves partial occlusion of a coronary artery, whereas STEMI involves complete occlusion

14. a — Physical activity, emotional stress, or cold weather

15. b — Nitroglycerin

16. b — The prognosis of death and ischemic events in ACS patients

17. b — The risk of in-hospital mortality

18. b — Severe chest pain with radiation, nausea, and sweating

19. a — Significant ST-segment elevation in ≥2 contiguous leads

20. b — Myocardial necrosis

21. a — The pain occurs at rest or with minimal exertion in UA

22. b — MONA (Morphine, Oxygen, Nitroglycerin, Antithrombotic therapy)

23. b — To reduce myocardial oxygen demand

24. a — Coronary Angiography followed by PCI (Percutaneous Coronary Intervention)

25. a — Coronary Artery Bypass Grafting (CABG)

26. a — V-A ECMO
كمان بيتستخدم كدعم IABP خلي بالك إن الـ V-A ECMO. الملف اختار ميكانيكي في الـ ACS.

27. a — Impella

28. b — Improves blood flow to the heart and reduces workload during PCI

29. c — ST-segment depression or T-wave inversion

30. a — It is a classification of heart failure severity, which impacts mortality risk

31. b — Risk of in-hospital mortality

32. b — Thrombolytics

33. b — Atherosclerotic plaque rupture and thrombus formation

34. b — It provides pain and anxiety relief

35. b — Prevent thrombus formation by inhibiting platelet aggregation

36. b — Elevated troponin levels

37. b — Transient reversible ischemia

38. b — Low blood pressure with signs of organ dysfunction

39. c — Proceed with coronary angiography and revascularization

N1. c — NSTEMI
ST depression/T inversion + raised troponin = NSTEMI. Same ECG with normal troponin = unstable angina.

N2. a, b, c — Unstable angina; NSTEMI; STEMI
Stable angina is chronic coronary disease, not ACS.

N3. b — Troponin I/T
Troponin rises 3–4 hrs, peaks 24 hrs, stays up 7–14 days.

N4. a — ≤ 90 minutes
If PCI is not available, give fibrinolytics with door-to-needle ≤ 30 minutes.

N5. a, b, c — Systolic BP > 90 mmHg; No recent phosphodiesterase inhibitor (sildenafil); No right ventricular infarction
Nitrates can cause severe hypotension in these situations.

N6. a — Aspirin 162–325 mg
Chewing speeds absorption; dual antiplatelet therapy adds clopidogrel/ticagrelor.

N7. a, b, c, e — Previous hemorrhagic stroke; Active internal bleeding; Suspected aortic dissection; Recent major surgery or trauma
Age alone is not a contraindication.

N8. b — Reducing myocardial oxygen demand
Avoid in heart failure, bradycardia, hypotension or cardiogenic shock.

N9. d — Cardiogenic shock
Class I none, II crackles/S3, III pulmonary edema, IV cardiogenic shock.

N10. a, b, c — Relief of chest pain; Resolution of ST elevation; Reperfusion arrhythmias
Accelerated idioventricular rhythm is a typical reperfusion arrhythmia.

N11. b — Continuous ECG monitoring with IV access and pain relief
Life-threatening arrhythmias (VF) are most common early.

N12. a — Atypical symptoms such as dyspnea, fatigue or epigastric pain
Silent or atypical MI is common in these groups.